Algorithmic Impact Assessment Results

Version: 0.10.0

Project Details

1. Name of Respondent
Jeff Hughes

2. Job Title
Senior Data Scientist

3. Department
Employment and Social Development (Department of)

4. Branch
CDOB

5. Project Title
Record of Employments (ROE) Comments Assessment

6. Project Phase
Implementation [Points: 0]

7. Please provide a project description:
The Chief Data Officer Branch (CDOB), Employment Insurance (EI) Benefits Delivery Services (BDS) and Innovation, Information and Technology Branch (IITB) are implementing an artificial intelligence (AI)/machine learning solution that will interpret and assess free text comments captured by employers when records of employment (ROE) are issued. Based on very specific business rules, the AI will assess and predict simple actions (i.e. save or ignore comments, predict a different Reason for Separation [RFS]). The actions taken following an AI prediction will be saved within the internal interpretative notes used in the process of calculating EI claims. The original document (Record of Employment [ROE]) will not be modified.

About The System

8. Please check which of the following capabilities apply to your system.
Text and speech analysis: Analyzing large data sets to recognize, process, and tag text, speech, voice, and make recommendations based on the tagging

Section 1: Impact Level : 1

Current Score: 31

Raw Impact Score: 37

Mitigation Score: 37

Section 2: Requirements Specific to Impact Level 1

Peer review

None

Gender-based Analysis Plus

None

Notice

None

Human-in-the-loop for decisions

Decisions may be rendered without direct human involvement.

Explanation

In addition to any applicable legal requirement, ensure that a meaningful explanation is published for common decision results. The explanation must provide a general description of:

- the role of the system in the decision-making process;
- input data, its source and method of collection;
- the criteria used to evaluate input data and the operations applied to process it;
- the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision; and
- the principal factors behind a decision.

Explanations must also inform clients of relevant recourse options, where appropriate.

Descriptions must be made available in plain language through the Algorithmic Impact Assessment and discoverable via a departmental website.

Training

None

IT and business continuity management

None

Approval for the system to operate

None

Other requirements

The Directive on Automated Decision-Making also includes other requirements that must be met for all impact levels.

Directive on Automated Decision-Making

Contact your institution's ATIP office to discuss the requirement for a Privacy Impact Assessment as per the Directive on Privacy Impact Assessment.

Section 3: Questions and Answers

Section 3.1: Impact Questions and Answers

Reasons for Automation

1. What is motivating your team to introduce automation into this decision-making process? (Check all that apply)
Lower transaction costs of an existing program
Use innovative approaches

2. What client needs will the system address and how will this system meet them? If possible, describe how client needs have been identified.
The system will assist in meeting service standards for benefit delivery by reducing the need for manual inspection of text comments in ROE forms. The system will be able to handle simple cases, leaving the more complex cases for human agents. Handling simple cases reduces the workload, freeing up agents' time for other aspects of EI benefits delivery, which may result in faster service delivery.

3. Please describe any public benefits the system is expected to have.
As the system will free up the time of human agents to focus on other aspects of EI benefits delivery, the public may benefit from faster service delivery, which has a general economic benefit. The cost savings of the system, allowing for more effective stewardship of public funds, is also a public benefit.

4. How effective will the system likely be in meeting client needs?
Very effective [Points: +0]

5. Please describe any improvements, benefits, or advantages you expect from using an automated system. This could include relevant program indicators and performance targets.
This system will be limited in scope to the examination of text comments in one field of the ROE form. Manual inspection of this field is time-consuming and yet often yields no relevant information, or information that is redundant with other areas of the ROE (Reason for Separation [RFS] section). We estimate that this AI system will be able to eliminate about 30-40% of work items related to manual inspection of text comments. This would work out, very roughly, to a cost savings of about $200k to $300k per year.

6. Please describe how you will ensure that the system is confined to addressing the client needs identified above.
The system only takes the text comment and the Reason for Separation (RFS) and Detail (RFS-DTL) codes as input; it does not examine any other portion of the ROE. There are checks in place to ensure that more complex cases are left untouched, and to ensure that the model is confident enough in its prediction before any action is taken. The model is designed to operate similarly to how human agents process ROE text comments, so the scope of the model is limited and designed to fit within the existing workflow, resulting in no changes to EI eligibility criteria or decision-making processes.

7. Please describe any trade-offs between client interests and program objectives that you have considered during the design of the project.
Client interests and program objectives are largely aligned in this context, in that both are interested in ensuring decisions are made quickly and correctly. The AI model will not be making decisions about eligibility, only about information contained in one portion of the ROE form. It is also set up to limit negative consequences of model errors; consideration was given to the full EI workflow with respect to when human intervention occurs. The system cannot make an error that would result in benefits erroneously being denied, without human intervention later in the process. Thus, the system is intended to improve the speed of service delivery, and minimize risk of error that would negatively impact clients.

8. Have alternative non-automated processes been considered?
Yes [Points: +0]

9. If non-automated processes were considered, why was automation identified as the preferred option?
The open text field in the ROE form is necessary for the provision of important additional information to provide context for an EI claim; removing it is not an option. Thus, it either needs to be reviewed manually, an automated system must be used, or some combination of the two. The preferred option, combining an automated system and human review, will reduce the workload for simple cases so human agents can spend their time on more complex cases.

10. What would be the consequence of not deploying the system?
Service delivery cannot achieve performance targets

Risk Profile

11. Is the project within an area of intense public scrutiny (e.g. because of privacy concerns) and/or frequent litigation?
No [Points: +0]

12. Are clients in this line of business particularly vulnerable?
No [Points: +0]

13. Are stakes of the decisions very high?
No [Points: +0]

14. Will this project have major impacts on staff, either in terms of their numbers or their roles?
No [Points: +0]

15. Will the use of the system create or exacerbate barriers for persons with disabilities?
No [Points: +0]

Project Authority

16. Will you require new policy authority for this project?
No [Points: +0]

About the Algorithm

17. The algorithm used will be a (trade) secret
No [Points: +0]

18. The algorithmic process will be difficult to interpret or to explain
Yes [Points: +3]

About the Decision

19. Please describe the decision(s) that will be automated.
The model will only take action on comments related to simple decisions, which are based on current procedures used by agents. These decisions involve determining if the comment contains information relevant to the Reason for Separation (RFS), and adjusting the RFS if necessary. When the AI interpretation is used in a decision, it will replace a decision currently made by a human that requires minimal judgement and/or discretion. However, a portion of the decisions (ones which cannot be assessed by AI, or more complex decisions) will continue to be made by humans.

20. Does the decision pertain to any of the categories below (check all that apply):
Social assistance (employment insurance, disability claims) [Points: +1]

Impact Assessment

21. Which of the following best describes the type of automation you are planning?
Partial automation (the system will contribute to administrative decision-making by supporting an officer through assessments, recommendations, intermediate decisions, or other outputs) [Points: +2]

22. Please describe the role of the system in the decision-making process.
When the AI system takes actions, it will adjust the Reason for Separation (RFS) in accordance with information in the text comment. This RFS code is used as part of the administrative decision-making process for determining eligibility for EI benefits. The AI system itself does not make any decisions about EI claims. In cases where the AI system does not take action, the text comment will be left for a human agent to inspect, and no changes will be made to the RFS.

23. Will the system be making decisions or assessments that require judgement or discretion?
Yes [Points: +4]

24. Please describe the criteria used to evaluate client data and the operations applied to process it.
The system evaluates the text information provided in the open comment field of the ROE, and attempts to classify it with respect to any information it may contain regarding a Reason for Separation (RFS). This system is a machine learning model that was trained on past ROE comments, to learn patterns associated with RFS information. The criteria involved include whether there is any information present about a reason for separation, and whether the existing RFS code provided on the ROE form is consistent with that information. No other information on the ROE is used in the evaluation.

25. Please describe the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision.
The system produces a prediction about the RFS information contained in the comment (if any). Additional business logic determines whether the RFS code should be adjusted to be in line with the information in the comment. The administrative decisions made regarding EI eligibility are made, in part, on the basis of the RFS code, for which information is provided on the EI Program's website. The criteria for EI eligibility will not change as a result of the automated system.

26. Will the system perform an assessment or other operation that would not otherwise be completed by a human?
No [Points: +0]

27. Is the system used by a different part of the organization than the ones who developed it?
Yes [Points: +4]

28. Are the impacts resulting from the decision reversible?
Reversible [Points: +1]

29. How long will impacts from the decision last?
Impacts are most likely to be brief [Points: +1]

30. Please describe why the impacts resulting from the decision are as per selected option above.
Information received on an ROE can normally be used on an EI claim for a period of one year, excluding certain exceptions where the period is extended and possible appeals. The impacts are reversible because the system is only addressing Reason for Separation (RFS) information in ROE text comments; administration and fact-finding steps from Service Canada agents later in the process could result in further adjustments to the RFS. The system does not make decisions about EI eligibility.

31. The impacts that the decision will have on the rights or freedoms of individuals will likely be:
Little to no impact [Points: +1]

32. Please describe why the impacts resulting from the decision are as per selected option above.
As the prediction made by the AI is used in support of a larger eligibility decisions, and is reversible, the impacts resulting from the decisions have little to no impact on the rights or freedoms of the individual applying for EI benefits.

33. The impacts that the decision will have on the equality, dignity, privacy, and autonomy of individuals will likely be:
Little to no impact [Points: +1]

34. Please describe why the impacts resulting from the decision are as per selected option above.
The system is designed to use minimal information in order to respect privacy and reduce the risk of unequal treatment. In addition, a fairness assessment was undertaken to evaluate the model, to ensure that the actions taken by the model are fair and unbiased across a range of sociodemographic groups.

35. The impacts that the decision will have on the health and well-being of individuals will likely be:
Little to no impact [Points: +1]

36. Please describe why the impacts resulting from the decision are as per selected option above.
The AI decision has no relation, hence no impacts, to the health and well-being of individuals.

37. The impacts that the decision will have on the economic interests of individuals will likely be:
Little to no impact [Points: +1]

38. Please describe why the impacts resulting from the decision are as per selected option above.
As it is completely reversible, the AI decision may only result in minor impacts to the speed at which the claim can be processed, if additional verification is required on a specific decision made. In the vast majority of cases, the processing speed will be unchanged or increased.

39. The impacts that the decision will have on the ongoing sustainability of an environmental ecosystem, will likely be:
Little to no impact [Points: +1]

40. Please describe why the impacts resulting from the decision are as per selected option above.
The decision will have very little impacts, if any, on the sustainability of an environmental ecosystem.

About the Data - A. Data Source

41. Will the Automated Decision System use personal information as input data?
Yes [Points: +4]

42. Have you verified that the use of personal information is limited to only what is directly related to delivering a program or service?
Yes [Points: +0]

43. Is the personal information of individuals being used in a decision-making process that directly affects those individuals?
No [Points: +0]

44. Have you verified if the system is using personal information in a way that is consistent with: (a) the current Personal Information Banks (PIBs) and Privacy Impact Assessments (PIAs) of your programs or (b) planned or implemented modifications to the PIBs or PIAs that take new uses and processes into account?
Yes [Points: +0]

45. Please list relevant PIB Bank Numbers.
ESDC PPU 385

46. What is the highest security classification of the input data used by the system? (Select one)
Protected B / Protected C [Points: +3]

47. Who controls the data?
Federal government [Points: +1]

48. Will the system use data from multiple different sources?
No [Points: +0]

49. Will the system require input data from an Internet- or telephony-connected device? (e.g. Internet of Things, sensor)
No [Points: +0]

50. Will the system interface with other IT systems?
Yes [Points: +4]

51. Who collected the data used for training the system?
Your institution [Points: +1]

52. Who collected the input data used by the system?
Your institution [Points: +1]

53. Please describe the input data collected and used by the system, its source, and method of collection.
The input data is provided by Record of Employment (ROE) forms submitted to ESDC by employers. The data used for training the system was collected from the administrative database which stores these ROE forms when they are submitted.

About the Data - B. Type of Data

54. Will the system require the analysis of unstructured data to render a recommendation or a decision?
Yes [Points: 0]

55. What types of unstructured data? (Check all that apply)
Audio and text files [Points: +2]

Section 3.2: Mitigation Questions and Answers

Consultations

1. Internal Stakeholders (federal institutions, including the federal public service)
Yes [Points: +1]

2. Which Internal Stakeholders have you engaged?
Strategic Policy and Planning
Data Governance
Program Policy
Legal Services
Access to Information and Privacy Office
TBS Office of the Chief Information Officer
Digital Policy

3. External Stakeholders (groups in other sectors or jurisdictions)
No [Points: +0]

De-Risking and Mitigation Measures - Data Quality

4. Do you have documented processes in place to test datasets against biases and other unexpected outcomes? This could include experience in applying frameworks, methods, guidelines or other assessment tools.
Yes [Points: +2]

5. Is this information publicly available?
Yes [Points: +1]

6. Have you developed a process to document how data quality issues were resolved during the design process?
Yes [Points: +1]

7. Is this information publicly available?
No [Points: +0]

8. Have you undertaken a Gender Based Analysis Plus of the data?
Yes [Points: +1]

9. Is this information publicly available?
No [Points: +0]

10. Have you assigned accountability in your institution for the design, development, maintenance, and improvement of the system?
Yes [Points: +2]

11. Do you have a documented process to manage the risk that outdated or unreliable data is used to make an automated decision?
Yes [Points: +2]

12. Is this information publicly available?
No [Points: +0]

13. Is the data used for this system posted on the Open Government Portal?
No [Points: +0]

De-Risking and Mitigation Measures - Procedural Fairness

14. Does the audit trail identify the authority or delegated authority identified in legislation?
Yes [Points: +1]

15. Does the system provide an audit trail that records all the recommendations or decisions made by the system?
Yes [Points: +2]

16. Are all key decision points identifiable in the audit trail?
Yes [Points: +2]

17. Are all key decision points within the automated system's logic linked to the relevant legislation, policy or procedures?

Yes [Points: +1]

18. Do you maintain a current and up to date log detailing all of the changes made to the model and the system?
Yes [Points: +2]

19. Does the system's audit trail indicate all of the decision points made by the system?
Yes [Points: +1]

20. Can the audit trail generated by the system be used to help generate a notification of the decision (including a statement of reasons or other notifications) where required?
Yes [Points: +1]

21. Does the audit trail identify precisely which version of the system was used for each decision it supports?
Yes [Points: +2]

22. Does the audit trail show who an authorized decision-maker is?
Yes [Points: +1]

23. Is the system able to produce reasons for its decisions or recommendations when required?
No [Points: +0]

24. Is there a process in place to grant, monitor, and revoke access permission to the system?
Yes [Points: +1]

25. Is there a mechanism to capture feedback by users of the system?
Yes [Points: +1]

26. Is there a recourse process established for clients that wish to challenge the decision?
Yes [Points: +2]

27. Does the system enable human override of system decisions?
Yes [Points: +2]

28. Is there a process in place to log the instances when overrides were performed?
Yes [Points: +1]

29. Does the system's audit trail include change control processes to record modifications to the system's operation or performance?
Yes [Points: +2]

30. Have you prepared a concept case to the Government of Canada Enterprise Architecture Review Board?
No [Points: +0]

De-Risking and Mitigation Measures - Privacy

31. If your system uses or creates personal information, have you undertaken a Privacy Impact Assessment, or updated an existing one?
Yes [Points: +1]

32. Please indicate the following in your answer: Title and scope of the Privacy Impact

Assessment; How the automation project fits into the program; and Date of Privacy Impact Assessment completion or modification.
A Privacy Impact Assessment, entitled "Record of Employment-Comment Artificial Intelligence Model" was completed April 2021. The scope of the privacy assessment included the evaluation of the security and safeguards related to the technologies used, including encryption and access controls; the collection, use, and retention of personal information; accuracy and monitoring of the automated solution; accountability for the solution; and procedures related to individuals' right to access their own personal information and file complaints or inquiries about policies and practices related to handling of this information. The assessment was specifically focused on the implementation of the automated system, not on the broader EI program and existing processes.

33. Have you designed and built security and privacy into your systems from the concept stage of the project?
Yes [Points: +1]

34. Is the information used within a closed system (i.e. no connections to the Internet, Intranet or any other system)?
Yes [Points: +1]

35. If the sharing of personal information is involved, has an agreement or arrangement with appropriate safeguards been established?
Yes [Points: +1]

36. Will you de-identify any personal information used or created by the system at any point in the lifecycle?
Yes [Points: +1]

37. Please describe your de-identification method(s).
Data are de-identified when provided to the AI system by removing personal information such as name, Social Insurance Number, and employer information. Outputs are similarly stored in a de-identified manner as well. Outputs are retained for a minimum of 12 years to align with the existing ROE retention period, deleted only when the ROE itself is deleted.